AUTOMOBILE & ELECTRIC VEHICLES

TASK 3 — COMPLETED ASSIGNMENT

EV Charging Systems, Energy Management & Efficiency Analysis

Prepared from the uploaded Task 3 brief

Includes explanations, comparison tables, worked examples, simple diagrams, and an EV case-study framework.

Important scope note

The uploaded PDF gives the assignment theme, required learning topics, practice exercises, and final-submission checklist. It does not provide a numbered set of detailed questions, prescribed calculation values, or a marking rubric in the extracted pages. This report therefore answers the listed required topics and uses clearly labelled illustrative examples where numerical data are not specified.

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1. EV Charging Fundamentals: AC vs DC

An electric vehicle (EV) stores electrical energy in a traction battery. Charging equipment transfers energy from the electrical supply to the battery under controlled conditions. The charging system must manage voltage, current, temperature, safety interlocks, and communication between the vehicle and the charging equipment.

1.1 AC charging

In AC charging, alternating current from the grid or AC charging point reaches the vehicle. The vehicle's on-board charger (OBC) converts AC into the DC required by the battery and controls the charging process. AC charging is common at homes, workplaces, and destinations where vehicles can remain parked for several hours.

1.2 DC charging

In DC charging, the charging station contains power-conversion equipment that supplies controlled DC to the vehicle's battery system. The vehicle and charger communicate to establish safe voltage and current limits. Because the main AC-to-DC conversion is performed in the external charger, DC charging can deliver much higher power than many on-board chargers, subject to the vehicle's battery, thermal system, state of charge, and station rating.

AC CHARGING PATH	DC CHARGING PATH
Grid AC → AC EVSE → On-board charger (AC→DC) → Battery	Grid AC → External DC charger (AC→DC) → Battery
Vehicle OBC limits charging power	Station and vehicle jointly limit charging power

1.3 Main comparison

Feature	AC charging	DC charging
Where conversion occurs	Inside the vehicle's on-board charger	Inside the external charging station
Typical use	Home, workplace, overnight or destination charging	Public rapid/fast charging, especially during travel
Charging speed	Often lower; limited by OBC and supply	Often higher; limited by charger, vehicle and battery conditions
Installation cost/complexity	Usually simpler for lower-power installations	Generally more expensive and power-intensive
Battery safety/control	Vehicle manages battery charging through OBC/control system	Vehicle and DC charger coordinate voltage/current and safety limits

Why is DC charging usually faster? The external station can provide higher DC power than a typical on-board charger can process. However, a DC charger's headline rating does not guarantee that the vehicle will continuously accept that power. Charging commonly slows at higher state of charge, and battery temperature or vehicle limits can reduce power.

1.4 Home versus public charging

• Home charging: convenient and often cheaper per unit of energy, especially when charging overnight; installation must be properly rated and protected.

• Workplace/destination charging: useful while the vehicle is parked for hours; AC charging is often sufficient.

• Public DC fast charging: reduces travel-stop time but can cost more and depends on station availability, compatibility, and vehicle charging curve.

Safety note: use a compatible, properly installed charger and cable; do not use damaged equipment or improvised wiring. Installation and protection should follow applicable local electrical rules and manufacturer instructions.

2. EV Charging Standards and Connectors

Charging standards define physical connectors, electrical ratings, communication methods, and safety requirements. A connector that physically fits is not by itself proof that a vehicle and charger are electrically or digitally compatible. Standards vary by region and vehicle type.

Connector / system	General description	Typical context / note
Type 1 (SAE J1772)	AC connector commonly associated with single-phase charging	Historically common in North America and some other markets; vehicle/region dependent
Type 2 (IEC 62196-2)	AC connector supporting single- and three-phase configurations, depending on equipment	Widely used in Europe and many other markets
CCS (Combined Charging System)	Combines an AC interface with additional DC contacts; CCS1 and CCS2 differ in connector design	Used for DC fast charging in multiple markets; confirm the exact vehicle inlet
CHAdeMO	A DC fast-charging system with its own connector and communication protocol	Found on some vehicle models and charging networks; adoption varies by market
GB/T	Chinese national charging standards, with AC and DC connector specifications	Common in mainland China; compatibility must be checked for imported vehicles
Bharat AC-001 / DC-001	Indian charging specifications associated with earlier low-power AC and DC charging infrastructure	Legacy/earlier ecosystem; check current local rules and the vehicle's supported standard

2.1 CCS versus CHAdeMO

Both are DC fast-charging systems, but they use different connector designs and communication systems. A vehicle with a CCS inlet cannot normally connect directly to a CHAdeMO plug without a compatible, approved solution. Network coverage, vehicle support, and regional market adoption should be checked before a trip.

2.2 Charging compatibility checklist

• Check the vehicle inlet and the connector available at the station.

• Check whether the vehicle supports AC, DC, or both, and its maximum accepted power.

• Confirm the station's rated power and whether the cable is tethered or requires the user's cable.

• Check payment/authentication requirements, operating status, and any network access requirements.

• For imported vehicles, verify regional connector and voltage/frequency compatibility rather than assuming that an adapter solves every issue.

> The uploaded brief asks the learner to study Type 1 vs Type 2, CCS vs CHAdeMO, and Indian Bharat AC/DC standards. It does not prescribe a single standard for a specific vehicle or provide a connector-rating table; the summary above is general technical context.

3. Charging Power, Energy, Time and Losses

Power and energy are related but are not the same quantity. Confusing them is one of the mistakes explicitly highlighted in the Task 3 brief.

Quantity	Meaning	Unit	Example
Power	Rate at which energy is transferred or used	kilowatt (kW)	A 7 kW charger can transfer energy at a nominal rate of 7 kW
Energy	Amount of electrical energy stored or consumed	kilowatt-hour (kWh)	A 50 kWh battery has a nominal energy capacity of 50 kWh

3.1 Basic charging-time calculation

For a simplified estimate, charging time is:

> Time (hours) ≈ Energy to be added (kWh) ÷ Average charging power (kW)

Example 1 — adding 30 kWh at an average of 6 kW:

Time ≈ 30 kWh ÷ 6 kW = 5 hours.

This is an idealised estimate. Actual time can be longer because charging power varies, the vehicle may limit the power, the battery-management system may taper power, and energy is lost in conversion and battery heating/cooling.

3.2 Including charging efficiency

If the battery must receive 30 kWh and the overall wall-to-battery efficiency is assumed to be 90%, the grid must supply approximately:

> Grid energy = Battery energy ÷ Efficiency = 30 ÷ 0.90 = 33.33 kWh.

If grid-side power were a constant 6 kW, the corresponding idealised time would be 33.33 ÷ 6 = 5.56 hours (about 5 h 33 min). In real charging, use average power over the session rather than assuming constant power.

3.3 Charging losses

- Power-electronics losses in the charger and on-board charger.
- Cable and connection resistance, which causes heat.
- Battery internal losses and energy used by battery thermal management.
- Vehicle systems that remain active during charging.

A useful efficiency definition is: Charging efficiency (%) = (energy stored in battery ÷ energy drawn from grid) × 100%. The boundary used for measurement should be stated, because different definitions may include or exclude auxiliary loads.

3.4 Why charging slows after about 80%

Many lithium-ion battery systems accept high power at lower and middle states of charge, then reduce current as the battery approaches its upper voltage limit. This tapering helps control voltage, heat, and degradation risk. The exact taper point and shape depend on the vehicle, battery chemistry, temperature, charger, and state of health; 80% is a useful rule of thumb, not a universal hard boundary.

3.5 Factors that change charging time

- Battery capacity and the amount of energy needed.
- Station power rating and available supply capacity.
- Vehicle’s maximum AC input / DC fast-charging acceptance.
- Battery state of charge and charging curve.

- Battery temperature and thermal-management strategy.
- Conversion losses and other auxiliary energy use.

4. EV Efficiency and Range Analysis

Energy efficiency indicates how much distance an EV travels for a given amount of energy. The Task 3 brief specifically identifies km/kWh and Wh/km as key measures.

4.1 Two common efficiency units

Efficiency in km/kWh = distance travelled (km) ÷ energy consumed (kWh). A larger number means more distance per unit of energy.

Consumption in Wh/km = energy consumed (kWh) × 1,000 ÷ distance travelled (km). A smaller number means less energy used per kilometre.

> These measures are reciprocal in scale: km/kWh ≈ 1,000 ÷ Wh/km. For example, 5 km/kWh corresponds to about 200 Wh/km.

4.2 Worked efficiency example

Suppose an EV travels 180 km and uses 36 kWh from its battery during that trip.

Efficiency = 180 ÷ 36 = 5 km/kWh.

Consumption = (36 × 1,000) ÷ 180 = 200 Wh/km.

4.3 Estimating range

> Estimated range (km) ≈ usable battery energy (kWh) × efficiency (km/kWh)

If an EV has 45 kWh of usable energy and achieves 5 km/kWh, estimated range = 45 × 5 = 225 km. This is an estimate, not a guarantee; usable energy, driving conditions and efficiency can vary.

4.4 Factors affecting real-world range

Factor	Effect on range
Speed and driving style	High speeds and rapid acceleration generally increase energy consumption; smoother driving can improve efficiency.
Traffic and route	Stop-start traffic can increase auxiliary and acceleration energy, while regenerative braking can recover part of the kinetic energy; terrain and congestion matter.
Temperature	Very hot or cold conditions can require cabin conditioning and battery thermal management; cold batteries may also limit regenerative braking and fast charging.
Heating / air conditioning	Cabin climate control consumes electrical energy and can reduce the energy available for driving.
Tyres and load	Low tyre pressure, extra mass and aerodynamic drag increase energy use.
Battery condition	Ageing and degradation can reduce usable energy; system limits and state of health matter.

4.5 Claimed range versus real-world range

A claimed range is obtained under a defined test procedure and set of assumptions. A real journey can differ because speed, ambient temperature, route elevation, traffic, tyres, payload, HVAC use, and driving style differ from the test. When comparing models, compare figures measured under the same test cycle where possible, and clearly label the cycle (for example, WLTP) rather than mixing test-cycle figures with independent real-world results.

Range comparison should be treated as an energy calculation, not just a battery-size comparison: a larger battery may provide more range, but a less efficient vehicle can use more energy per kilometre.

5. Regenerative Braking

Regenerative braking allows an EV’s traction motor to operate as a generator during deceleration. Instead of converting all of the vehicle’s kinetic energy into heat at the friction brakes, part of that energy is converted into electrical energy and returned to the battery.

Energy flow during regeneration
Vehicle kinetic energy → Motor/generator → Power electronics → Battery (stored electrical energy)

5.1 Benefits

• Recovers some energy that would otherwise be dissipated as heat during braking.

• Can improve overall efficiency and driving range, especially on routes with repeated deceleration.

• Can reduce friction-brake use and associated wear in suitable conditions.

5.2 Practical limitations

• Regeneration is limited by battery state of charge; a nearly full battery may accept little or no regenerative energy.

• Battery temperature and battery-management limits can reduce available regeneration.

• At low speeds, during emergency braking, or when strong braking is required, friction brakes may still be needed.

• Tyre grip, stability control and motor/inverter limits constrain the amount of regenerative torque.

• On steady high-speed routes with few braking events, less energy is available to recover.

Regeneration does not create energy: it recovers part of the vehicle’s kinetic energy, with losses in the motor, inverter, battery and drivetrain. It cannot recover all energy used to accelerate because conversion is not perfectly efficient and aerodynamic/rolling resistance also dissipate energy.

6. Real-EV Case Studies and Comparison

The uploaded brief specifically names the Tata Nexon EV, Tesla Model 3, and Hyundai Kona Electric for case-study research. It asks learners to investigate battery capacity, AC/DC charging time, real-world range, and efficiency. The supplied PDF does not state model years, variants, battery sizes, official figures, or a test cycle. Because those details differ by market and model year, the following is a structured case-study analysis rather than a claim that one fixed specification applies to every version.

6.1 Case study A — Tata Nexon EV

Vehicle context: a compact electric SUV sold in India in different versions over time. For a completed model-specific submission, select one exact model year and battery variant from the manufacturer's official specification sheet.

Analysis to record:

• Battery capacity: record the stated gross or usable capacity and identify which is reported.

• AC charging: record the on-board AC charging rating and the time for a defined charge window (for example, 10–100%); note that the actual time depends on the supply and losses.

• DC charging: record the peak DC acceptance and manufacturer-stated time for a defined state-of-charge interval.

• Real-world range: use a documented road test or clearly described owner/test data, not an unlabeled estimate.

• Efficiency: calculate km/kWh or Wh/km from energy and distance measured under the same conditions.

6.2 Case study B — Tesla Model 3

Vehicle context: a mid-size electric sedan offered in multiple battery, motor, and regional configurations. Charging connector, charging network access, battery capacity and stated range vary by market and model year.

Key analysis: compare the vehicle's supported AC charging rate with its DC fast-charging curve; record the test cycle used for official range; then compare it with a real-world road test that states speed, temperature, route and state-of-charge window. Peak DC power alone is not enough to estimate a full-session charge time.

6.3 Case study C — Hyundai Kona Electric

Vehicle context: a compact electric SUV available in different battery versions and generations. A meaningful comparison must specify the exact variant, model year, market, and charging standard.

Key analysis: identify battery capacity and AC/DC limits; compare the manufacturer's defined charging-time window with observed time; calculate consumption from measured trip energy and distance; and explain differences between claimed and real-world range.

6.4 Comparison table for final researched figures

Parameter	Tata Nexon EV	Tesla Model 3	Hyundai Kona Electric
Exact model year / variant	To be specified from chosen source	To be specified from chosen source	To be specified from chosen source
Battery capacity (kWh)	Insert official value; note gross/usable	Insert official value; note gross/usable	Insert official value; note gross/usable
AC charge rating / time	Insert for stated charge window	Insert for stated charge window	Insert for stated charge window
DC peak / time window	Insert official value and window	Insert official value and window	Insert official value and window
Claimed range / test cycle	Insert official range and cycle	Insert official range and cycle	Insert official range and cycle
Real-world range	Insert cited test result and conditions	Insert cited test result and conditions	Insert cited test result and conditions
Efficiency (km/kWh or Wh/km)	Calculate from a consistent dataset	Calculate from a consistent dataset	Calculate from a consistent dataset

How to complete the table responsibly: use official manufacturer documents for specifications and stated charging times; use a reputable, documented road test for real-world performance; write the publication date and URL in the reference list; and do not combine figures from different variants without labelling them. The source brief lists the three vehicles but does not provide their specifications or ask for a specific model year, so unsupported numerical values have deliberately not been invented.

7. Charging Infrastructure and Energy Management

Charging infrastructure includes the grid connection, distribution equipment, protection devices, EV supply equipment (EVSE), connectors/cables, communication and payment systems, and the vehicle's charging hardware. Reliable charging depends on the complete system, not only the charger's power rating.

7.1 Main infrastructure elements

Element	Purpose
Grid / service connection	Provides electrical supply at the required voltage and capacity.
Distribution board and protection	Provides correctly rated circuits and appropriate overcurrent, earth-leakage and other protection required by local code.
EVSE / charging station	Controls safe energy transfer, interlocks and communication; DC stations also perform high-power AC-to-DC conversion.
Connector and cable	Transfers power and provides the required signal/communication paths.
Backend/network	May manage authentication, billing, monitoring, fault reporting and charger availability.
Vehicle charging system	Checks conditions and controls battery current, voltage, temperature and charging limits.

7.2 Energy management strategies

• Scheduled charging: charge when the vehicle is parked and, where suitable, during lower-cost or lower-demand periods.

• Load management: coordinate multiple chargers so the combined demand stays within the site's electrical capacity.

• Smart charging: adjust charging power or timing in response to tariffs, grid demand, renewable generation, or user requirements.

• Renewable integration: when available, align charging with solar generation while maintaining user needs and system safety.

• Monitoring: track kWh delivered, session duration, faults and charger availability to improve utilisation and diagnose problems.

7.3 PCB-mounted relays in EV charging systems

The uploaded PDF includes the phrase "Next Generation PCB-mounted Relays In EV Charging Systems" but does not supply a detailed question or technical specifications for a particular relay. In general, relays are electrically controlled switching devices. In EVSE control circuits, appropriately rated relays may switch control or auxiliary circuits; power contactors are commonly used where high current and isolation requirements demand them. A component must be selected according to voltage, current, inrush, insulation, temperature rise, endurance, creepage/clearance, and applicable safety standards. A small PCB relay should never be assumed suitable for switching traction-battery or high-power charging current unless its ratings and system design explicitly permit it.

A relay's coil/control circuit and its switched contacts are distinct. Control logic can command a relay, but safety-rated interlocks, monitoring, and correctly rated switching/protection hardware are required to prevent energisation when a connector is unsafe or a fault is present.

8. Worked Practice Exercises

These examples answer the practice themes in the uploaded brief. Values are illustrative because the PDF does not specify numerical exercise inputs.

Exercise 1 — Compare AC and DC charging

A vehicle needs 24 kWh added. Assume average AC charging power at the battery is 6 kW and average DC charging power at the battery is 30 kW.

AC time ≈ 24 ÷ 6 = 4 hours. DC time ≈ 24 ÷ 30 = 0.8 hours = 48 minutes.

Interpretation: under these simplified assumptions, DC is five times the average power and the time is one-fifth. Real sessions may take longer due to power tapering, temperature, vehicle limits and losses.

Exercise 2 — Charging different battery sizes

Energy to add	Average power	Idealised time
20 kWh	5 kW	20 ÷ 5 = 4 h
40 kWh	10 kW	40 ÷ 10 = 4 h
60 kWh	30 kW	60 ÷ 30 = 2 h

The examples show that time depends on both energy to be added and average power. They do not model charging curves or losses.

Exercise 3 — Calculate efficiency and estimated range

An EV consumes 42 kWh to travel 210 km. Efficiency = 210 ÷ 42 = 5 km/kWh. Consumption = 42,000 ÷ 210 = 200 Wh/km. If its usable battery energy is 50 kWh and the same efficiency is maintained, estimated range = 50 × 5 = 250 km.

Exercise 4 — Explain a range difference

An EV’s measured consumption increases from 160 Wh/km to 200 Wh/km under a different set of conditions. Range from the same 50 kWh usable battery changes from 50,000 ÷ 160 = 312.5 km to 50,000 ÷ 200 = 250 km. The estimated reduction is 62.5 km, or 20%. Possible reasons include higher speed, cold weather, cabin heating, low tyre pressure, extra load or a hilly route. This is a mathematical illustration, not a claim about a particular EV.

Exercise 5 — Charging infrastructure

Before using a public charger, verify connector compatibility, station power, the vehicle’s accepted charging power, operating status, payment/access requirements, and any restrictions on charging duration. For a site with several chargers, load management can prevent simultaneous charging demand from exceeding the site’s electrical capacity.

Exercise 6 — Regenerative braking

Explain the energy path: vehicle kinetic energy → motor acting as generator → power electronics → battery. State at least two limitations: a full/cold battery may accept less energy, and friction braking is still needed for some braking events. Energy recovery is partial, not complete.

9. Common Mistakes and Final Submission Checklist

9.1 Common mistakes to avoid

• Mixing up kW and kWh: kW measures power; kWh measures energy.

• Using peak power for the entire session: charging power changes with state of charge and conditions.

• Ignoring losses: grid energy drawn is generally greater than energy stored in the battery.

• Assuming claimed range equals real range: test-cycle results and actual journeys have different conditions.

• Comparing unlike variants: model year, battery version, region and test cycle must be identified.

• Assuming connector shape guarantees compatibility: standards and communication requirements matter.

• Overstating regenerative braking: it recovers only part of the available kinetic energy.

9.2 Final submission checklist from the brief

Required item in source brief	Included in this report	Status / note
Charging system explanation	Sections 1 and 7	Covered
Charging standards table	Section 2	General comparison; verify current local details for a selected vehicle
Charging calculations	Sections 3 and 8	Worked illustrative examples; input values were not supplied
Range analysis	Section 4 and Exercise 3–4	Formulas and worked examples included
Case study report	Section 6	Framework and comparison table; model-specific numerical data not in source PDF

The PDF also recommends learning resources and tools, including YouTube topic searches, NPTEL, Coursera audit mode, a laptop, internet access, a calculator/Excel, and optional spreadsheet or presentation tools. These are learning aids rather than technical answers and are not reproduced as a separate bibliography of every link.

10. Conclusion and Source / Scope Notes

EV charging performance depends on the interaction between the supply, charging station, vehicle charging hardware, battery-management system, battery condition, and charging state. AC charging relies on the vehicle's on-board charger for AC-to-DC conversion, whereas DC fast charging performs that conversion externally and can generally deliver higher power. Charging time must be calculated from energy added and average power, with allowance for losses and charging-power tapering.

Efficiency and range are linked: km/kWh measures distance per unit of energy, while Wh/km measures energy per distance. Real-world range is affected by speed, temperature, terrain, traffic, tyres, load, climate control and battery condition. Regenerative braking improves efficiency by returning some kinetic energy to the battery, but conversion losses and practical limits prevent complete recovery.

Source basis

Primary source: the user-uploaded "Automobile & Electric Vehicles – Task 3" PDF. Its extracted content identifies the task as "EV Charging Systems, Energy Management & Efficiency Analysis," names the learning topics and three case-study vehicles, and lists the required final submission items. It does not provide detailed numbered questions, numerical inputs for calculations, model-year/variant selections, a grading rubric, or full specifications for the case-study vehicles.

Scope and assumptions

• The explanations, equations and illustrative numerical examples in this report are general engineering knowledge used to answer the topics named in the brief.

• All numerical exercises are explicitly illustrative and should not be presented as manufacturer specifications.

• The case-study table intentionally leaves model-specific figures to be filled from dated, traceable official specifications and road-test sources. This avoids inventing values or mixing different variants.

• Charging standards and vehicle specifications can change over time and by market. Confirm current details for the exact model and location before final submission.

Suggested reference practice for completing the case-study figures

• Manufacturer specification sheets and owner manuals for the exact model year and variant.

• Official charging-network or manufacturer pages for charging connector and power compatibility.

• A reputable road test that states route, speed, temperature, test method and energy measurement.

• Record each source's title, publisher, publication/update date, and URL, and cite each figure in the final version.

End of completed assignment draft